

THE CHINESE UNIVERSITY OF HONG KONG
Department of Mathematics
SAYT1114 Number Theory and Cryptography (Summer 2026)
Tutorial 1 Supplementary Notes - On Proofs

- If you have any questions, feel free to contact us in the WhatsApp group!
- Unless otherwise stated, all letters stand for an integer. Furthermore, we shall agree that

$$\mathbb{N} = \{1, 2, 3, \dots\},$$

not including zero.

In everyday language, ‘proof’ often means convincing someone beyond a reasonable doubt. In mathematics, a proof is a logical argument that demonstrates a statement is always true, without exception. Unlike science, which relies on experiments and observations, math relies on strict, step-by-step logic. A million examples showing a pattern works do not make it a mathematical fact; a proof does. In this note, we will guide you through common proof techniques.

1 Direct proof

A typical mathematical statement looks like this:

Example 1. If m and n are even, then $m - n$ is also even.

When we perform direct proofs, we usually follow this outline:

1. Assume the premise holds.
2. Perform logical/ algebraic manipulations.
3. Arrive at the desired conclusion.

Let’s see how example 1 can be proved in the above framework.

1. The premise here reads

- m and n are even.

In mathematical language, we translate it into

- Let $m = 2k$ and $n = 2\ell$.

2. Our goal is to prove $m - n$ is even, so we should compute $m - n$:

$$m - n = 2k - 2\ell = 2(k - \ell).$$

3. Now we know $m - n$ is indeed even, which completes our proof.

The formal proof is organized as follows:

Proof. Since m and n are even, let $m = 2k$ and $n = 2\ell$. Then we have

$$m - n = 2k - 2\ell = 2(k - \ell).$$

Therefore $m - n$ is even. ||

2 Case-by-case consideration

Sometimes, instead of proving a general statement at once, we may want to break it down into cases, and attack them case by case. See the following example:

Example 2. $n(n + 1)$ is even.

Proof. n is either even or odd, so we discuss them separately:

- **Case I:** Suppose n is even, say $n = 2k$. Then we have

$$n(n + 1) = 2k(2k + 1)$$

which is even.

- **Case II:** Suppose n is odd, say $n = 2k + 1$. Then we have

$$n(n + 1) = (2k + 1)(2k + 2) = 2(k + 1)(2k + 1)$$

which is even.

In any case, the statement holds. Therefore, the proof is completed. ||

3 Proof by contrapositive

Proofs can get very complicated if we go straight ahead. A little twist might make things easier. Suppose we are given the following statement:

Example 3. If n^2 is even, then n is even.

If we attempt to prove it directly, we will get things like this:

Since n^2 is even, let $n^2 = 2k$. Then $n = \sqrt{2k}$... What's going on!?

We can see direct proof would not work easily, so let's prove the following equivalent statement:

(*) If n is NOT even, then n^2 is NOT even.

(*) is called the **contrapositive** of the original statement. To persuade you of the equivalence, let's think of the following examples:

- Original statement: If it is raining outside, then the sportsground must be wet.
- Contrapositive: If the sportsground is NOT wet, then it is NOT raining outside.
- **Wrong statement:** If the sportsground is wet, then it is raining outside. (Why wrong? It might be my dog, Chad, who was too happy and spilled his water bowl in the sportsground, making it wet!)
- **Wrong statement:** If it is NOT raining outside, then the sportsground must NOT be wet.

If you agree with the fact that example 3 and (*) are equivalent, then you are ready for the following proof:

Proof. Suppose, for the sake of contrapositive, that n is not even. Then n is odd, say $n = 2k + 1$. Then we have

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Therefore n^2 is odd, i.e. n^2 is not even. By contrapositive, if n^2 is even, then n is even. ||

Some words are underlined. They provide signals to the readers that we are using proof by contrapositive. You are highly recommended to include such indicators in your proofs, so that your tutors can follow your proofs more easily.

4 Proof by contradiction

As we have said, ‘proofs can get very complicated if we go straight ahead; a little twist might make things easier.’ Here we provide the second approach to introduce such a plot twist, namely proof by contradiction. Suppose we are given the following statement:

Example 4. If $a, b, c \in \mathbb{N}$ satisfy the Pythagorean equation

$$a^2 + b^2 = c^2,$$

then a and b cannot be both odd.

To prove ‘if A then B’ using proof by contradiction, we would assume B is false first, and then show that this assumption leads to a logical absurdity or impossibility (what your TAs refer to as ‘EXPLOSION!!!’).



Let’s see how it works:

Proof. Suppose, on the contrary, that a and b are both odd. Let

$$a = 2m + 1, \quad b = 2n + 1.$$

Then we have

$$\begin{aligned} c^2 = a^2 + b^2 &= (2m + 1)^2 + (2n + 1)^2 \\ &= 4m^2 + 4m + 1 + 4n^2 + 4n + 1 \\ &= 2(2m^2 + 2m + 2n^2 + 2n + 1) \end{aligned}$$

which means c^2 is even. By example 3, c is also even. Let $c = 2k$. Then

$$4k^2 = (2k)^2 = c^2 = 2(2m^2 + 2m + 2n^2 + 2n + 1).$$

Cancelling the factor 2 on both sides, we have

$$2k^2 = 2m^2 + 2m + 2n^2 + 2n + 1 = 2(m^2 + m + n^2 + n) + 1$$

The left-hand side is even, but the right-hand side is odd. **This is nonsense.**

||

When you work with proof by contradiction, it is necessary to indicate that you make the world explode. This is the function of the **blue words**. You may use other wordings like ‘contradiction arises’ or ‘this is absurd’ or the symbol $\rightarrow\leftarrow$. No matter what style you prefer, such indicators are absolutely necessary!

On the other hand, the underlined words also provide signals to the readers that we are using proof by contradiction. You are highly recommended to include such indicators in your proofs, so that your tutors can follow your proofs more easily.

5 If and only if

Sometimes, you may see the word ‘**if and only if**’. For example:

Example 5. Let $k \in \mathbb{N}$ and p be a prime. Suppose $\alpha, \beta \in \mathbb{C}$ are the roots of the equation

$$x^2 - (k+2)x + p = 0.$$

Then $\alpha, \beta \in \mathbb{Z}$ if and only if $k = p - 1$.

Here we can see the statement

$$\alpha, \beta \in \mathbb{Z} \text{ if and only if } k = p - 1.$$

We can also present it symbolically:

$$\alpha, \beta \in \mathbb{Z} \iff k = p - 1.$$

This is the ‘zipped’ version of the following two statements:

	Symbolically	Verbally
$(\implies)$ part	$\alpha, \beta \in \mathbb{Z} \implies k = p - 1$	If $\alpha, \beta \in \mathbb{Z}$, then $k = p - 1$.
$(\impliedby)$ part	$\alpha, \beta \in \mathbb{Z} \impliedby k = p - 1$	If $k = p - 1$, then $\alpha, \beta \in \mathbb{Z}$.

To prove example 5, we need to prove two statements: the $(\implies)$ part and the $(\impliedby)$ part.

Proof.

$(\implies)$ part: Suppose $\alpha, \beta \in \mathbb{Z}$. By the sum of roots and product of roots formulas, we have

$$\alpha + \beta = k + 2 \quad \text{and} \quad \alpha\beta = p.$$

This means α and β are positive. Moreover, as $\alpha\beta = p$ where p is a prime, we have

$$\begin{cases} \alpha = p \\ \beta = 1 \end{cases} \quad \text{or} \quad \begin{cases} \alpha = 1 \\ \beta = p \end{cases}.$$

In either case, we have $\alpha + \beta = p + 1$. Therefore we have

$$p + 1 = k + 2 \implies k = p - 1.$$

$(\impliedby)$ part: Suppose $k = p - 1$. Then the equation becomes

$$x^2 - (p+1)x + p = 0.$$

Solving, we have

$$x = 1 \quad \text{or} \quad x = p.$$

Since α, β are the roots of the equation, it follows that $\alpha, \beta \in \mathbb{Z}$. ||

You are reminded that the notation $\iff$ should be used with care. Most of the time, implications only work one way but not the other. For example, think about what is wrong with the following implication?

$$x^2 = 1 \iff x = 1.$$

6 Proof by induction

Suppose we have a long chain of dominoes.



We want to know whether every domino will topple. We know two things:

1. The first domino always topples.
2. If the k th domino topples, then the $(k + 1)$ th domino will also topple.

Can we conclude that every domino topples?

- By 1, we know the first domino topples.
- Since the first domino topples, by 2 we know the second domino topples.
- Since the second domino topples, by 2 we know the third domino topples.

Continuing in this way, we can show that the fourth domino topples, the fifth domino topples, and so on. What about the 114514th domino? The answer is still yes, which follows from applying 2 for 114513 times. But can we conclude every domino in this long chain will eventually topple? The answer is yes, something we can verify using an important proof technique called **mathematical induction**.

Theorem 6.1 (Principle of mathematical induction). Let $\mathcal{P}(n)$ be a proposition about integers n . Suppose that:

1. $\mathcal{P}(1)$ is true.
2. if $\mathcal{P}(k)$ is true for some positive integer k , then $\mathcal{P}(k + 1)$ is also true.

Then $\mathcal{P}(n)$ is true for all positive integers n .

To prove a proposition using the principle of mathematical induction, follow the following quintet:

1. Define the proposition $\mathcal{P}(n)$.
2. Prove that $\mathcal{P}(1)$ is true.
3. Assume $\mathcal{P}(k)$ is true for some positive integer k .
4. Prove that $\mathcal{P}(k + 1)$ is true.
5. Conclude that $\mathcal{P}(n)$ is true for all positive integers n .

Example 6. For all $n \in \mathbb{N}$, it holds

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Proof. Let $\mathcal{P}(n) : \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$.

When $n = 1$,

$$\begin{aligned}\text{LHS} &= \sum_{k=1}^1 k^2 = 1^2 = 1 \\ \text{RHS} &= \frac{n(n+1)(2n+1)}{6} = \frac{1 \times 2 \times 3}{6} = 1 = \text{LHS}\end{aligned}$$

Therefore $\mathcal{P}(1)$ is true.

Assume $\mathcal{P}(m)$ is true for some positive integer m , i.e.

$$\sum_{k=1}^m k^2 = \frac{m(m+1)(2m+1)}{6}.$$

When $n = m + 1$,

$$\begin{aligned}\text{LHS} &= \sum_{k=1}^{m+1} k^2 \\ &= \sum_{k=1}^m k^2 + (m+1)^2 \\ &= \frac{m(m+1)(2m+1)}{6} + (m+1)^2 \\ &= \frac{m+1}{6} [m(2m+1) + 6(m+1)] \\ &= \frac{m+1}{6} [2m^2 + 7m + 6] \\ &= \frac{(m+1)(m+2)(2m+3)}{6} \\ &= \text{RHS}\end{aligned}$$

Therefore $\mathcal{P}(m+1)$ is true given that $\mathcal{P}(m)$ is true.

By mathematical induction, $\mathcal{P}(n)$ is true for all $n \in \mathbb{N}$. ||

7 Disproofs

Sometimes you might be given statements like this:

- All mathematics students are single.

How can you refute this kind of statement? It is easy: just find a mathematics student who has a girlfriend or a boyfriend. This is called **disproof by counterexample**. You may expect to see some questions like this in your exam:

Example 7. Determine whether the statement is true or false. If the statement is true, please provide a proof. If the statement is false, please provide a counterexample.

(#) If $n \in \mathbb{N}$, then $f(n) = n^2 - n + 41$ is a prime.

To prove a general statement like (#) is false, we don't need a lengthy argument. We only need to find a single counterexample where the statement fails, just like the mathematics student who has a girlfriend or a boyfriend in the above introduction.

Proof. Note that

$$f(41) = 41^2 - 41 + 41 = 41^2$$

which is composite. Therefore, statement (#) is false. ||

8 Final remarks

A famous educational YouTuber encourages students to prove things by working backwards, namely, assume the conclusion you need, and aim at a universal truth. This is probably the worst thing you can do. Let's see an example:

Example 8 (Wrong example). Prove that $-1 = 1$.

In his philosophy, we work like this:

$$\begin{aligned} -1 &= 1 \\ (-1)^2 &= 1^2 \\ 1 &= 1 \end{aligned}$$

which is a universal truth. Then we would present the formal proof like this:

Wrong proof.

$$\begin{aligned} 1 &= 1 \\ (-1)^2 &= 1^2 \\ -1 &= 1 \end{aligned}$$
||

Obviously, this is nonsense. After all, universal truths can easily be deduced from total nonsense. Students should never work like this.

Exercise. Determine whether the statement is true or false. If the statement is true, please provide a proof. If the statement is false, please provide a counterexample. You are not supposed to use anything after **Proposition 1** in the lecture notes.

1. If $a \mid b$ and $b \mid c$, then $a \mid c$.
2. $n(n+1)(n+2)$ is divisible by 3.
3. If $m+n$ is odd, then m or n is odd.
4. If mn is odd, then m and n are odd.
5. $17 \mid (9x+5y)$ if and only if $17 \mid (2x+3y)$.
6. $4^n - 1$ is divisible by 3 for all $n \in \mathbb{N}$.
7. If $a \mid b$, then $a^2 \mid b^3$.
8. If $a^2 \mid b^3$, then $a \mid b$.

Appendix: Set notations

We assume you are already familiar with basic set notation (which appeared on your admission test). Nonetheless, here is a quick review.

1. Any 'well-defined' collection of objects is called a **set**. The objects which make up a set will be called **elements** or **members** of the set. If an object x is an element of a set A , then we write $x \in A$, otherwise we write $x \notin A$. In saying that A is a well-defined collection of objects if given any object x , then either $x \in A$ or $x \notin A$ but not both.
2. A set A containing finite elements is called a **finite set**, otherwise, is an **infinite set**. Suppose A is a finite set. We may denote the set by putting its complete membership list between a pair of braces. For example,

$$A = \{1, 2, 3, 4\}.$$

But for infinite set or finite set containing a huge number of elements, it is impossible to list all its elements. So we use the **set-builder notation** to describe a set:

$$A = \{x : \mathcal{P}(x)\}.$$

Here x is only a symbol. $\mathcal{P}(x)$ is a compound statement of x , which is called the property of x . Sometimes we use $f(x)$ (a formula about x) instead of x at the symbolic part. For example,

$$A = \{n \mid n \text{ is an integer between 1 and 4}\} = \{1, 2, 3, 4\}$$

$$C = \{n^2 - n + 41 \mid n \text{ is an integer}\}$$

3. Some commonly used sets:
 - $\mathbb{N} = \{1, 2, 3, \dots\}$ is the set of all natural numbers. (Very few number theorists agree that 0 is a natural number.)
 - $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ is the set of all integers.
 - $\mathbb{Q} = \left\{\frac{p}{q} : p, q \in \mathbb{Z} \text{ where } q \neq 0\right\}$ is the set of all rational numbers.
 - $\mathbb{R}$ is the set of all real numbers.
 - $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}, i^2 = -1\}$ is the set of all complex numbers.
4. Suppose A and B are sets. B is a **subset** of A if $x \in B$ implies $x \in A$. In this case, we write $B \subset A$ or $A \supset B$. In addition, if $B \neq A$, then we write $B \subsetneq A$ or $A \supsetneq B$. Moreover, $A = B$ if and only if $A \subset B$ and $B \subset A$. Thus $A = \{1, 2, 3, 4\}$ and $B = \{1, 3, 2, 4, 4\}$ are equal.
5. Suppose A and B are sets. The **union** and the **intersection** of A and B are defined as follows:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$